Quantification: 536-63x_Hyp_RD_z08c1-2
======================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	19
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 19)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2669.84	2736	1499.86	px^2
major_axis_length_px	lacuna shape	99.0415	102.242	40.5286	px
minor_axis_length_px	lacuna shape	39.1288	39.2919	11.2588	px
aspect_ratio	lacuna shape	2.5275	2.5123	0.884	unitless
eccentricity	lacuna shape	0.8845	0.9174	0.0757	unitless
solidity	lacuna shape	0.7772	0.8492	0.1359	unitless
roots_count	headline	10.7895	10	5.7983	count
ring_length_r30_px	headline	312.053	305	169.268	px
ring_length_r60_px	reported	635.263	601	336.707	px
owned_length_px	ownership-dependent	848.373	842.269	719.363	px
edge_count	ownership-dependent	41.8947	38	34.3865	count
mean_edge_length_px	ownership-dependent	20.6999	19.753	6.5957	px
perimeter_px	normalised	303.303	274.735	131.113	px
ring_area_r30_px2	normalised	9484.95	10082	3265.38	px^2
ring_area_r60_px2	normalised	21315.1	22623	7346.24	px^2
in_frame_fraction_r30	normalised	0.9864	1	0.0448	unitless
in_frame_fraction_r60	normalised	0.9728	1	0.0801	unitless
ring_density_r30	normalised	0.0314059	0.031131	0.00866122	px^-1
ring_density_r60	normalised	0.0285455	0.0278216	0.00786244	px^-1
roots_per_100px_perimeter	normalised	3.4759	3.5057	0.9316	per 100 px
ring_attached_length_r30_px	network v2	279.263	272	163.882	px
ring_attached_length_r60_px	network v2	490.474	438	300.167	px
ring_length_w_r30_px	network v2	352.192	350.978	189.566	px
ring_length_w_r60_px	network v2	722.16	682.541	377.497	px
sholl_crossings_r10	network v2	7.5789	8	3.3219	count
sholl_crossings_r20	network v2	8.2632	9	4.1479	count
sholl_crossings_r30	network v2	8.4211	7	4.1542	count
loop_count	loops, report only	2.2632	1	3.4935	count
loop_edge_share	loops, report only	0.1344	0.1316	0.1195	share
ring_width_mean_r30_px	width	7.5527	7.3803	0.6112	px

Per field
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measure	value	units
analysed_area_px2	945450	px^2
total_skeleton_length_px	25555	px
canalicular_length_density_per_px	0.0270295	px^-1
junction_count	494	count
junction_density_per_px2	0.0005225	px^-2
lacuna_count	19	count

lacunae_per_px2	2.01e-05	px^-2
skeleton_component_count	362	count
mean_component_length_px	70.5939	px
median_component_length_px	24.5	px
field_length_density_w_per_px	0.0306956	px^-1
field_density_without_flagged_per_px	0.0256296	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	331	px
unattached_component_count	19	count
canalicular_length_density_attached_per_px	0.0270295	px^-1
field_length_density_with_unattached_per_px	0.0273795	px^-1
excluded_area_px2	52399	px^2
analysed_field_area_px2	996177	px^2
loop_count	53	count
loop_edge_share	0.1506	share
edge_count	1348	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7355	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25542	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2181	148.407	31.7971	4.6673	0.9768	0.5491	-0.2241	127.792	674.16	False
2	3003	146.448	39.2919	3.7272	0.9633	0.5868	-0.1342	219.3	634.066	False
3	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
4	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
5	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
6	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
7	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
8	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
9	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
10	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
11	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
12	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
13	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
14	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
15	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
16	584	33.791	23.1054	1.4625	0.7297	0.9125	-0.7852	793.788	813.11	False
17	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
18	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
19	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	21	392	645	542.245	38	14.2696	7.9442
2	14	325	728	943.122	56	16.8415	7.2425
3	20	457	794	1082.08	45	24.0461	7.528
4	19	723	1433	1324.53	81	16.3522	6.6187
5	6	202	455	726.612	31	23.4391	7.3169
6	10	325	580	1119.45	52	21.5278	7.3803
7	20	632	1263	3256.5	156	20.875	7.0906
8	4	90	197	79.1838	7	11.312	7.454
9	8	165	372	227.338	12	18.9448	6.9494
10	13	411	890	1092.82	61	17.9151	8.8041
11	10	286	601	967.637	42	23.039	8.2735
12	6	200	473	298.563	17	17.5626	7.9746
13	11	305	670	1297.54	48	27.0322	8.1069
14	10	349	860	1189.62	50	23.7924	7.3406
15	4	91	217	232.208	16	14.513	6.8986
16	2	96	145	82.9117	2	41.4558	7.9853
17	11	289	439	498.416	33	15.1035	7.1548
18	10	375	855	842.269	33	25.5233	8.5809
19	6	216	453	316.049	16	19.753	6.858

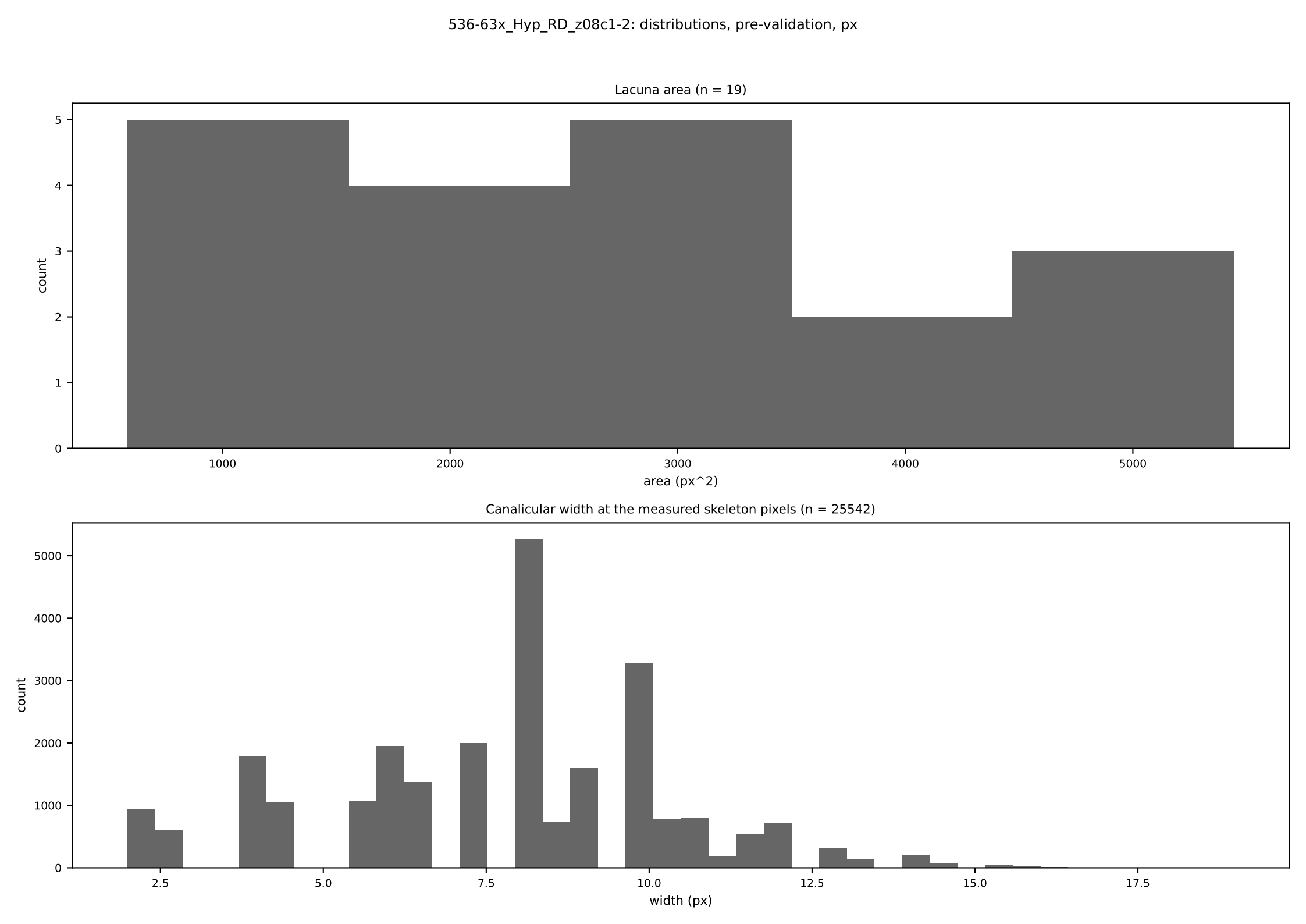
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	415.641	10082	20733	1	1	0.0388812	0.0311098	5.0524
2	495.973	10286	21637	1	1	0.0315963	0.0336461	2.8227
3	442.446	12800	26586	1	1	0.0357031	0.0298653	4.5203
4	514.931	14099	31464	1	1	0.0512802	0.0455441	3.6898
5	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
6	423.747	11743	23622	0.8152	0.7003	0.0276761	0.0245534	2.3599
7	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
8	156.16	5263	12719	1	1	0.0171005	0.0154886	2.5615
9	228.202	6707	15935	1	1	0.0246012	0.0233448	3.5057
10	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731
11	227.38	9187	22623	1	1	0.031131	0.0265659	4.3979
12	274.735	10452	25593	1	1	0.0191351	0.0184816	2.1839
13	250.693	9821	24082	1	1	0.0310559	0.0278216	4.3878
14	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
15	131.776	4471	9578	1	1	0.0203534	0.0226561	3.0355
16	102.847	3623	7408	1	1	0.0264974	0.0195734	1.9446
17	259.557	9006	18739	1	1	0.0320897	0.0234271	4.238
18	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
19	234.309	8189	17896	0.9258	0.8017	0.0263769	0.0253129	2.5607

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	342	438	441.747	736.24	8	12	10
2	313	588	369.563	829.24	8	8	7
3	428	723	517.546	907.98	10	10	12
4	694	1052	813.127	1613.81	14	11	16
5	202	435	228.924	526.002	6	3	5
6	325	552	368.291	663.198	9	10	7
7	563	1136	709.985	1430.87	13	16	16
8	76	76	102.326	221.995	4	0	2
9	150	205	184.196	422.818	3	5	5

10	371	812	458.635	1009.29	8	11	13
11	243	396	326.735	682.541	12	7	10
12	149	242	228.865	544.571	6	9	7
13	305	602	350.978	779.181	8	9	8
14	261	536	373.551	951.796	8	11	11
15	91	197	104.083	252.622	3	3	2
16	47	56	109.983	170.409	3	4	5
17	272	404	334.635	504.073	7	7	7
18	329	642	423.019	955.595	10	15	12
19	145	227	245.451	518.801	4	6	5


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
Canalicular width at the measured skeleton pixels (n = 25542)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5